A literature comparison matrix

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Keep the meaning beside the number. Preserve the missing fields.

Build a complete fictional literature matrix and justify comparability without forcing headline-score rankings.

Keep one habit: Matrix contract → different-task filled table → indoor/outdoor picture → missing-field probes → reconstruction protocol → independent metric/corpus mismatch.

MAIN LESSON · SECTION 1

Compare study meanings before comparing headline numbers

> Preserve task, metric, cases and missing fields in one matrix.

Learning goals

I can explain the **mechanism** and terminology of a literature comparison matrix.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve source versions, methods and evidence scope

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Replication and reproduction. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

My starting explanation

End goal

Explain a literature comparison matrix using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of a literature comparison matrix

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Define a matrix that retains differences and uncertainty

> A well-filled table explains why a direct ranking may be unavailable.

Begin with two percentages measuring different tasks

Imagine one study reports ninety percent classification accuracy and another eighty percent claim support for generated answers. Sorting 90 above 80 does not make the first tool better for the second task. A literature matrix keeps the meaning of each result beside it: question, data, unit, baseline, metric, split, resources and limitations. “Not reported” is useful information.

It is more honest than inventing a denominator or filling an unknown version from another study.

Define a literature comparison matrix

A **literature matrix** compares studies using aligned columns rather than sorting headline numbers with different meanings.

Complete study-field and comparison contract

Matrix rule	Complete fictional comparison protocol
Study identity	Keep exact source/version/locator when known; supplied examples have no real citation

Matrix rule	Complete fictional comparison protocol
Question and task	Name what each study evaluates before comparing outcomes
Cases and unit	Record dataset/population, sample denominator and experimental unit; mark absent details not reported
Baseline and metric	Keep reference method, outcome definition, cutoff and aggregation rule
Split and resources	Record training/evaluation separation, environment and resource measurements if supplied
Limitations	State known task/condition gaps and unknown methods
Comparison judgment	Aligned for a stated purpose / different known fields / insufficiently reported; no automatic rank
No invented counts	A percentage alone does not supply its raw numerator or denominator
Scope	Fictional matrix practice, not real literature synthesis, pooled meta-analysis or tool recommendation

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the main studies through a completed meaning-preserving matrix

> Unknown denominators stay unknown rather than becoming convenient fractions.

Original matrix-field procedure

Record question, data, unit, baseline, metric, split, resources and limitations. Mark incomparable fields rather than forcing a ranking.

Complete main task/metric/unknown-field matrix

Main matrix field	Study A supplied record	Study B supplied record
Task	Classification, described as easy	Generated answers, described as difficult
Reported metric/result	Accuracy90%	Claim support 80%
Dataset/corpus and version	Not reported	Not reported
Evaluation unit	Classification cases implied; exact definition not reported	Claim versus whole-answer aggregation not reported
Raw count/denominator	Not reported; do not invent 9/10	Not reported; do not invent 8/10
Baseline/model version	Not reported	Not reported
Split and resource conditions	Not reported	Not reported
Known comparison gap	Different task/outcome from B	Different task/outcome from A

Main matrix field	Study A supplied record	Study B supplied record
Valid conclusion	A’s headline percentage is numerically higher	No direct cross-task superiority follows

State the comparison still needed

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

A literature comparison matrix — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Study A reports accuracy 90% on easy classification; B reports claim support 80% on difficult generated answers.	Record question, data, unit, baseline, metric, split, resources and limitations. Mark incomparable fields rather than forcing a ranking.	Different tasks and metrics cannot establish A superior to B. Keep separate columns and identify a common evaluation needed.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict a matrix that hides indoor versus outdoor conditions

> Removing a column removes information, not the underlying difference.

Current task and response

- Study A reports accuracy 90% on easy classification; B reports claim support **80%** on difficult generated answers. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the missing-condition column

- Change: Remove the indoor/outdoor column.

▼ **Check after predicting**

Prediction: The matrix conceals an important condition difference.

Reason: A reader might now mistake unlike studies for a controlled head-to-head comparison.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read equal rates alongside different case conditions

> Keep denominators and missing alignment details visible.

Main different-task comparison and checked limit

Study A reports accuracy 90% on easy classification; B reports claim support **80%** on difficult generated answers.

Different tasks and metrics cannot establish A superior to B. Keep separate columns and identify a common evaluation needed.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate indoor/outdoor records and complete matrix

A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions.

Study S1: 8/10 correct on indoor photos. S2: 16/20 on outdoor photos. Both report **80%**.

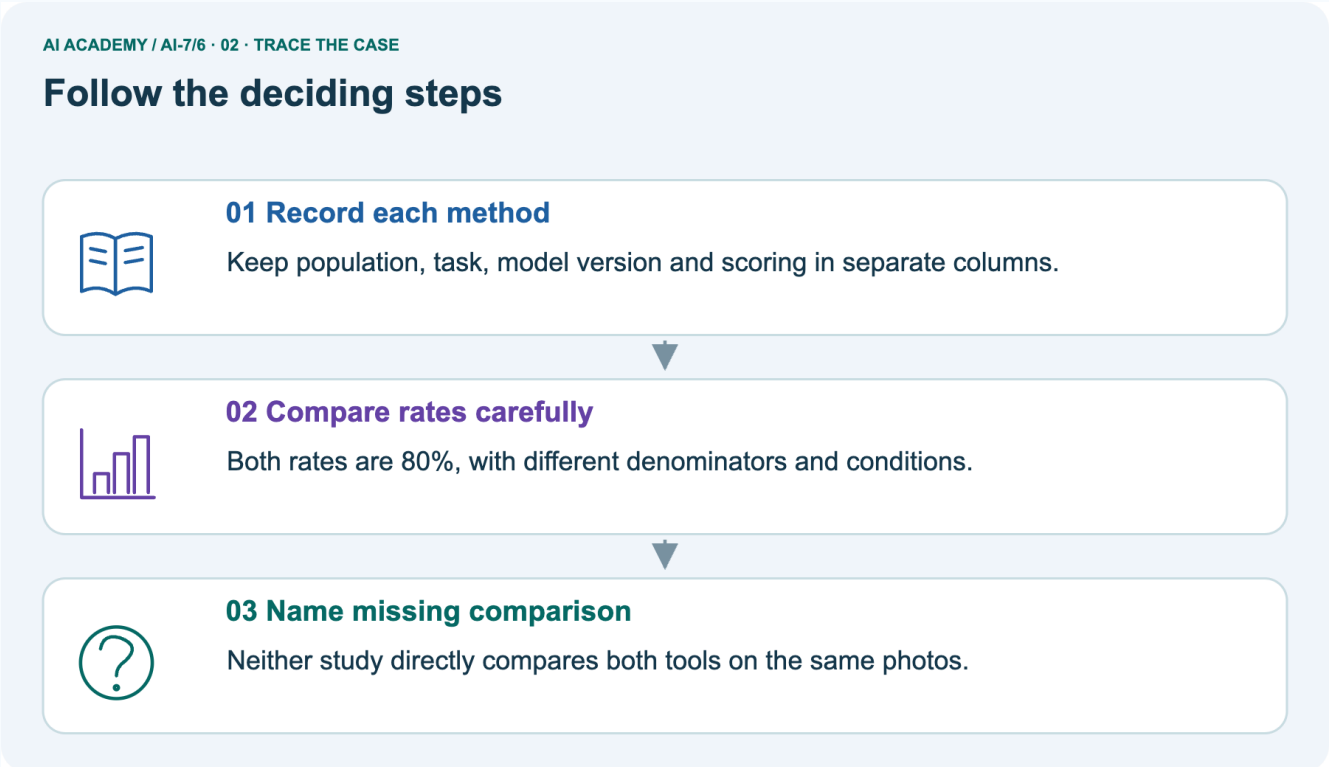
The picture is a separate photo-study comparison. S1 reports 8/10 correct on indoor photos; S2 reports 16/20 on outdoor photos. Both rates are 80%, but the samples, denominators and conditions differ. Neither supplied record directly evaluates both tools on the same photos. Equal percentages do not establish equivalent performance or identical uncertainty.

Removing the indoor/outdoor column makes the matrix less informative while leaving the original difference intact. To answer a shared application question, first define an appropriate common task, eligible data, scoring, versions and conditions; a matrix by itself does not run that evaluation.

Separate picture matrix field	S1	S2
Photo condition	Indoor	Outdoor
Correct / evaluated	8/10	16/20
Observed correctness rate	80%	80%
Exact photo IDs/population	Not reported	Not reported
Tool/model version and baseline	Not reported	Not reported

Separate picture matrix field	S1	S2
Scoring comparability	Correctness reported; exact label protocol not supplied	Correctness reported; exact label protocol not supplied
Split/resources	Not reported	Not reported
Direct same-photo A/B evaluation	Not supplied	Not supplied
Interpretation	Indoor supplied-set result	Outdoor supplied-set result; no direct equivalence/ranking established

Trace study methods, descriptive rates and missing comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record each method	Keep population, task, model version and scoring in separate columns.
2. Compare rates carefully	Both rates are 80%, with different denominators and conditions.
3. Name missing comparison	Neither study directly compares both tools on the same photos.

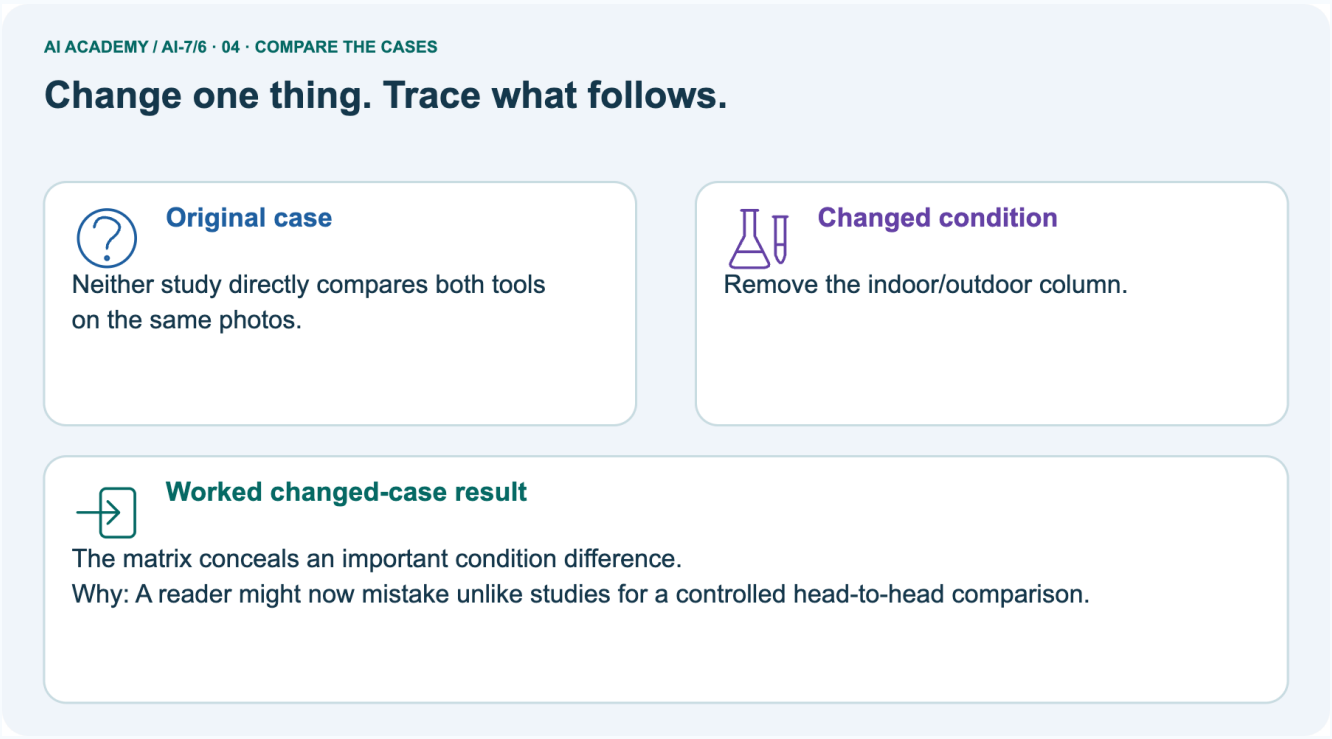
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **literature matrix** helps compare **evidence** without flattening differences. Equal percentages do not mean equivalent tasks, samples or methods.

Predict removal of indoor/outdoor column

- Change one thing: Remove the indoor/outdoor column. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain fictional-study scope and preserved conditions

- These fictional studies illustrate comparison practice; no real literature finding is asserted.
- Read the labelled picture: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the **literature matrix** preserve?

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: A literature comparison matrix

The facts we will use

AI ACADEMY / AI-7/6

Same rate, different study conditions

	Correct	Total	Condition
Study S1	8	10	Indoor
Study S2	16	20	Outdoor

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/6

A literature comparison matrix

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record each method

Keep population, task, model version and scoring in separate columns.



2. Compare rates carefully

Both rates are 80%, with different denominators and conditions.



3. Name missing comparison

Neither study directly compares both tools on the same photos.

CHANGE ONE THING

Remove the indoor/outdoor column.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit hidden differences, absent fields and forced rankings

> A comparison table should preserve reasons for caution rather than flatten them.

Core matrix qualification and failure analysis

A matrix documents comparison limits; it does not create missing common **evidence**.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete comparability and missing-field probes

Literature-matrix probe	Expected interpretation
90% and 80% use different tasks/metrics	Do not rank tools by the scalar numbers alone
Same80% on different samples	Equal observed rates, not established equivalence
Denominator absent	Mark not reported; no reconstructed raw count
Indoor/outdoor field removed	Important known condition hidden
Both sources omit a field	Unknown agreement, not proven matching conditions
Different metric/cutoff/corpus	No direct score ranking; align the intended comparison first
One study has more cases	More reported cases alone does not solve task or quality mismatch
Average incompatible percentages	No meaningful pooled result without a justified common estimand/protocol
Matrix completed	Evidence organization, not new experiment or causal synthesis

Guided repair and explanation

- A report omits this qualification: A matrix documents comparison limits; it does not create missing common **evidence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise explaining a comparison gap using named fields

> Propose what needs to be aligned rather than choosing the larger number.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 trace

Use workbook W2 for the connected example “A literature comparison matrix”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for A literature comparison matrix. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the **literature matrix** preserve?

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the selected recall/precision/corpus mismatch

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Complete a matrix reconstruction and comparability explanation

> Use the table to expose what is known, different and still missing.

Current different-task headline investigation

Study A reports accuracy 90% on easy classification; B reports claim support **80%** on difficult generated answers. Show the intermediate steps and give the conclusion with its scope.

Complete extraction and alignment protocol

Complete matrix-building investigation	Expected evidence
Extract main task and metric	Classification accuracy 90% versus generated-answer claim support 80%
Fill every other main field	Use not reported where no denominator/version/baseline/split/resource data exists
Write a justified main comparison	Different task and metric; no direct superiority ranking
Build separate S1/S2 rows	Indoor8/10=80%, outdoor 16/20=80%; retain condition and count
Hide then restore condition field	Explain which distinction becomes invisible, without changing the source evidence
Specify a revealing next comparison	Common application task, eligible judged data, scoring/aggregation, versions and conditions
Record scope	Supplied fictional-source inspection only; no real literature search or shared evaluation performed

Expected matrix and evidence-status limits

Use the complete filled matrices as worked references, then reconstruct their main fields from the supplied source statements. Distinguish a known mismatch from an unreported field. Explain why unknown/unknown is not evidence of equivalence. In the picture, compute the two 80% rates but keep their denominators and indoor/outdoor conditions.

No raw cases, actual paper identities or tool versions are supplied, so do not invent them. The exercise produces a comparison record and proposed common evaluation, not a ranking, new experiment or real literature finding.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

Try alone: recall at five versus precision at three on another corpus

> Keep metric, cutoff and data differences separate.

Complete selected independent task

A has recall@5 .8; B has precision@3 .9 on another corpus. Can you rank them?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

A has recall@5 .8; B has precision@3 .9 on another corpus. Can you rank them?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save source identity, aligned fields and unresolved comparisons

> Keep not-reported entries and source-specific limitations intact.

Project task and artifact

Add a labelled a literature comparison matrix **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of a literature comparison matrix.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual supplied-record inspection and remaining gaps

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain why the matrix supports a comparison or withholds a ranking

> Name the common evidence still needed.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of a literature comparison matrix

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Literature matrix	A structured comparison of study methods and results
Comparable metric	A measure with aligned definition and scope
Evidence gap	Information needed but not supplied

Use two terms to explain a step in this week's example.

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A matrix documents comparison limits; it does not create missing common **evidence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

A literature comparison matrix — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Study A reports accuracy 90% on easy classification; B reports claim support 80% on difficult generated answers.	Record question, data, unit, baseline, metric, split, resources and limitations. Mark incomparable fields rather than forcing a ranking.	Different tasks and metrics cannot establish A superior to B. Keep separate columns and identify a common evaluation needed.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions.

By the end, I can	How I will show it
Explain the mechanism and terminology of a literature comparison matrix	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions.
- Which labelled step matches this detail? Neither study directly compares both tools on the same photos.
- 1. Name missing comparison
- 2. Record each method
- 3. Compare rates carefully
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Name missing comparison
Neither study directly compares both tools on the same photos. A **literature matrix** helps compare **evidence** without flattening differences.

Equal percentages do not mean equivalent tasks, samples or methods.

Changed-case question and optional hint

- Changed-case practice: Remove the indoor/outdoor column. Which conclusion is supported by the supplied facts?
- 1. The matrix conceals an important condition difference.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The matrix conceals an important condition difference. A reader might now mistake unlike studies for a controlled head-to-head comparison.

Repair and retain the record

Review Replication and reproduction (ai-7/5). Goal: Explain the **mechanism** and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Review Finding primary research (ai-7/3). Goal: Explain the **mechanism** and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records.

Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work
</details>

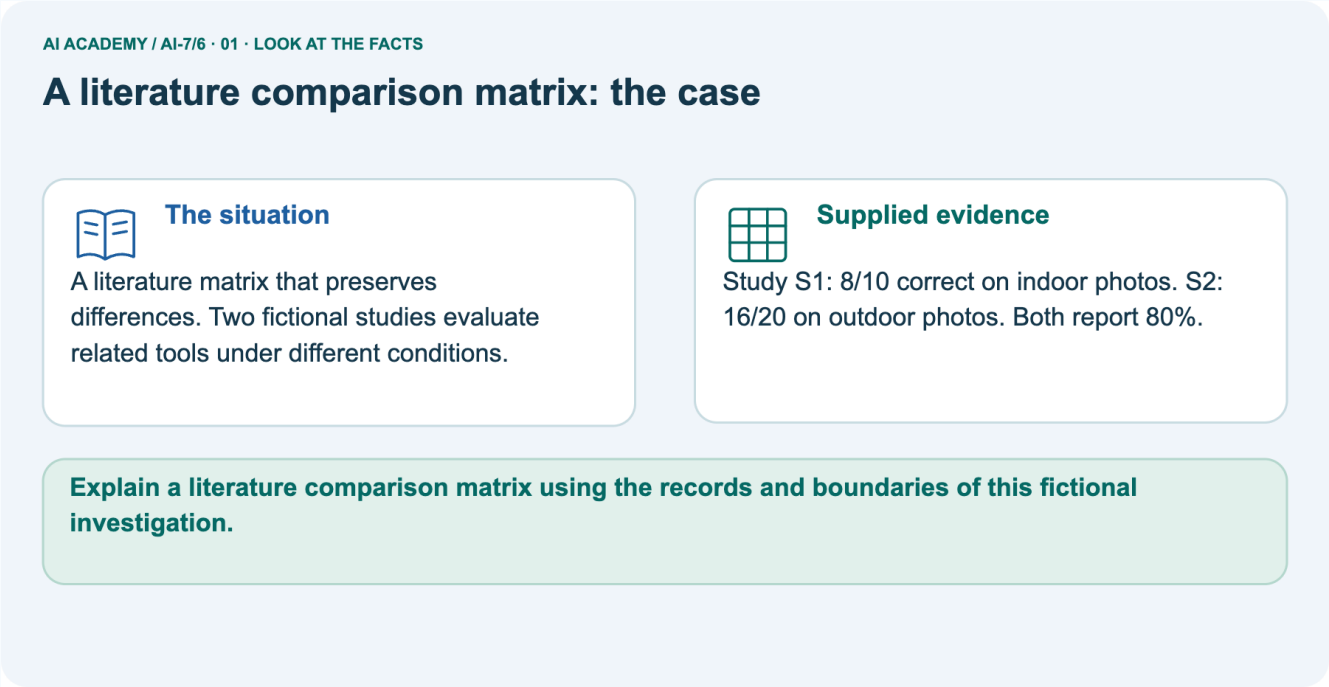
<details>
<summary>Reference · Six-picture concept journey</summary>
</details>

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. A literature comparison matrix: the case

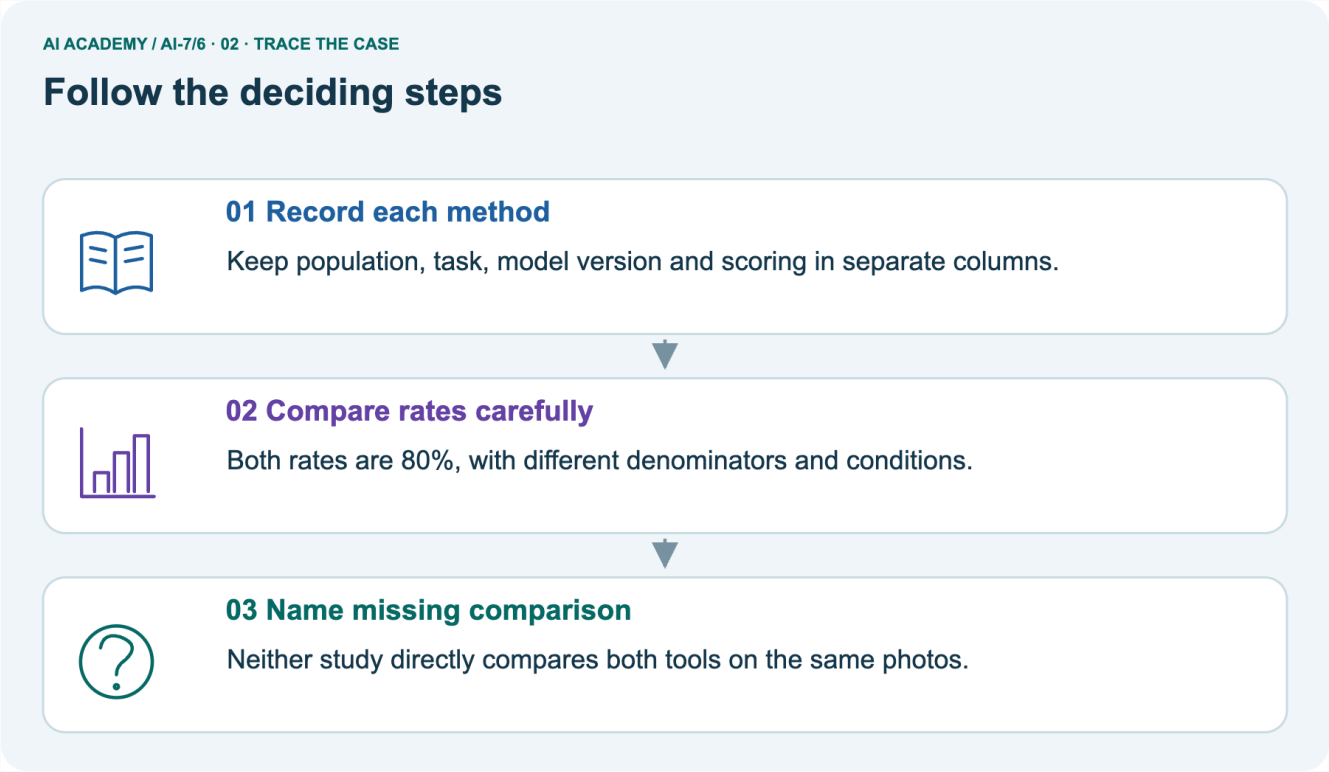


Follow the labelled picture with the matching explanation. Open the vector diagram

A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions. Supplied facts: Study S1: 8/10 correct on indoor photos. S2: 16/20 on outdoor photos. Both report **80%**.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

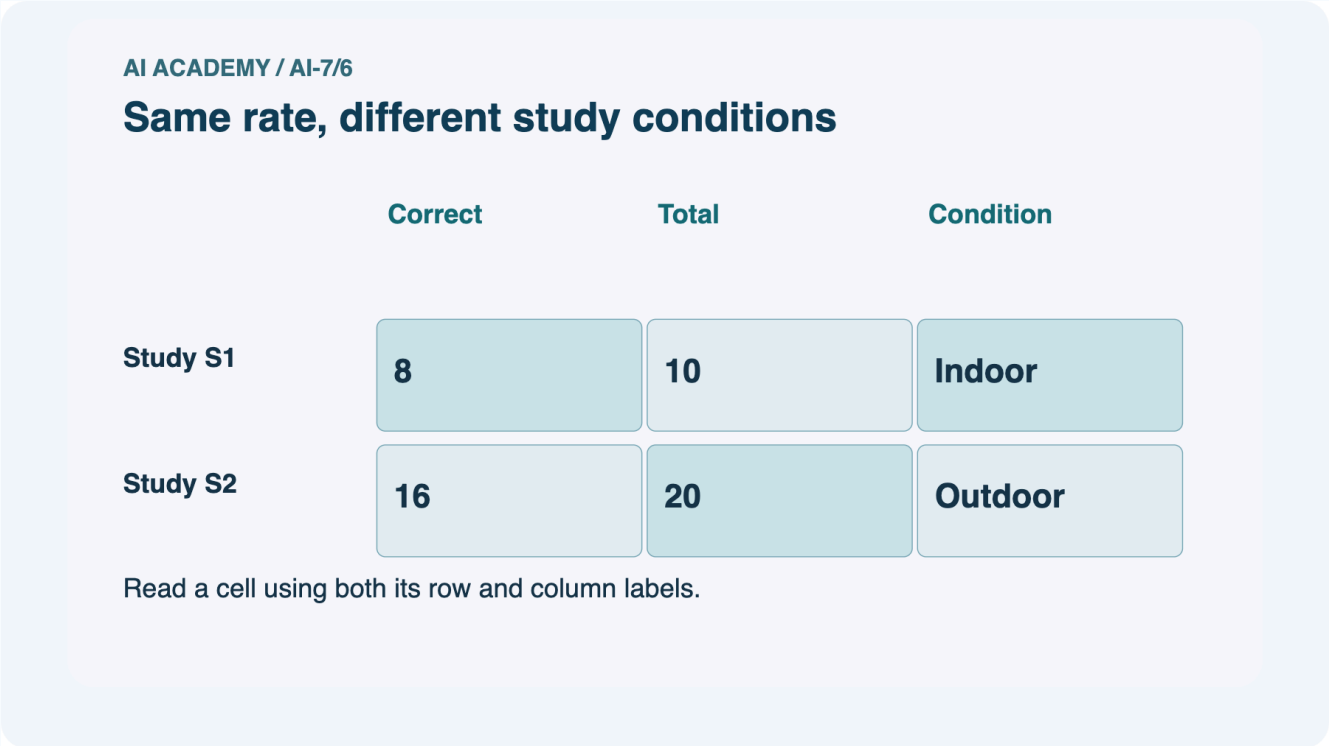


Follow the labelled picture with the matching explanation. Open the vector diagram

Record each method: Keep population, task, model version and scoring in separate columns.
→ Compare rates carefully: Both rates are **80%**, with different denominators and conditions.
→ Name missing comparison: Neither study directly compares both tools on the same photos.

Point to each numbered stage and explain which recorded **input** it uses.

03. Same rate, different study conditions

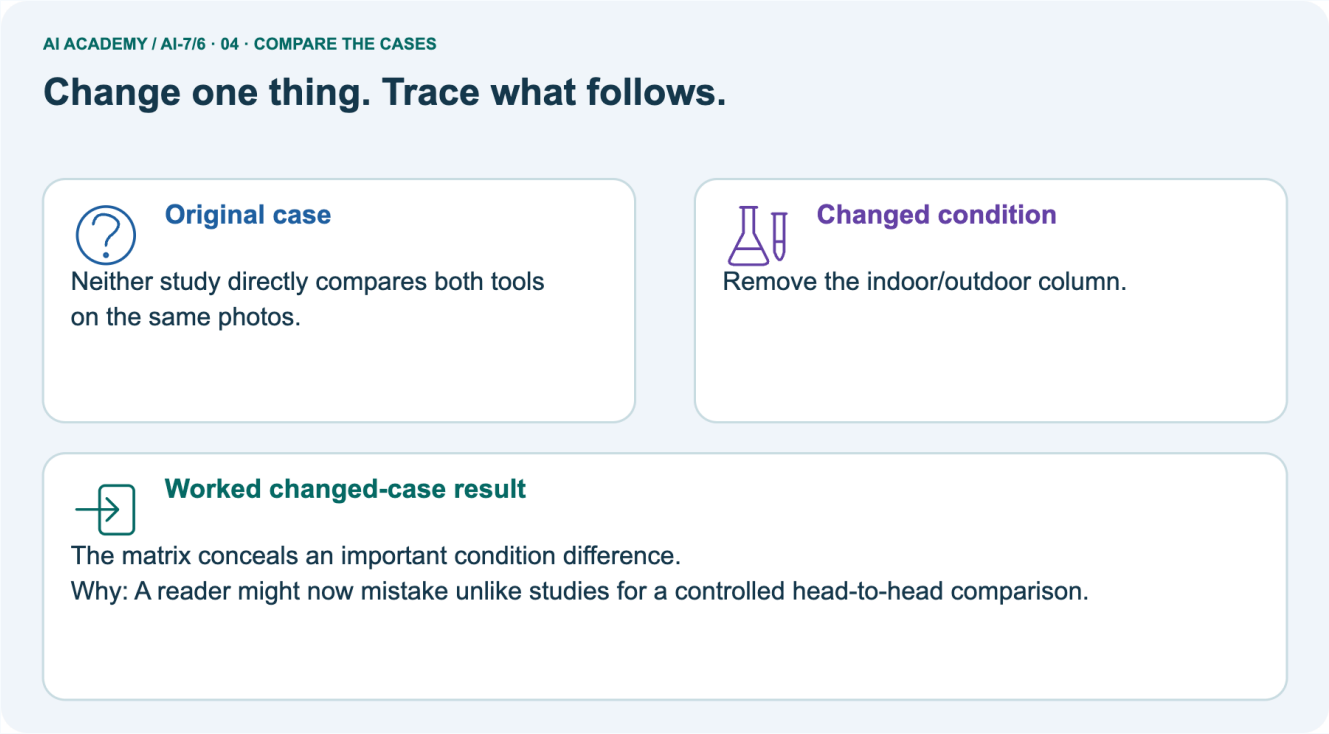


Follow the labelled picture with the matching explanation. Open the vector diagram

Same rate, different study conditions. Study S1: Correct = 8, Total = 10, Condition = Indoor. Study S2: Correct = 16, Total = 20, Condition = Outdoor.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



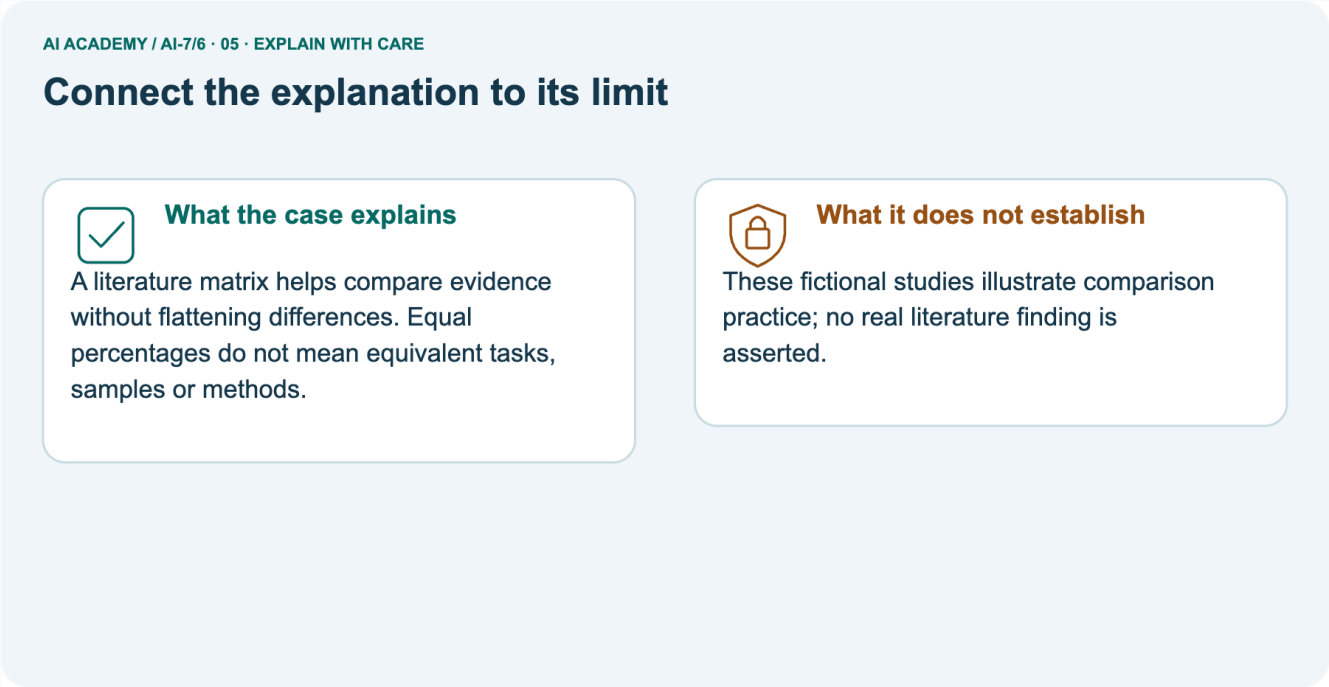
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Neither study directly compares both tools on the same photos. Changed condition: Remove the indoor/outdoor column. Expected result: The matrix conceals an important condition difference.

Reason: A reader might now mistake unlike studies for a controlled head-to-head comparison.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



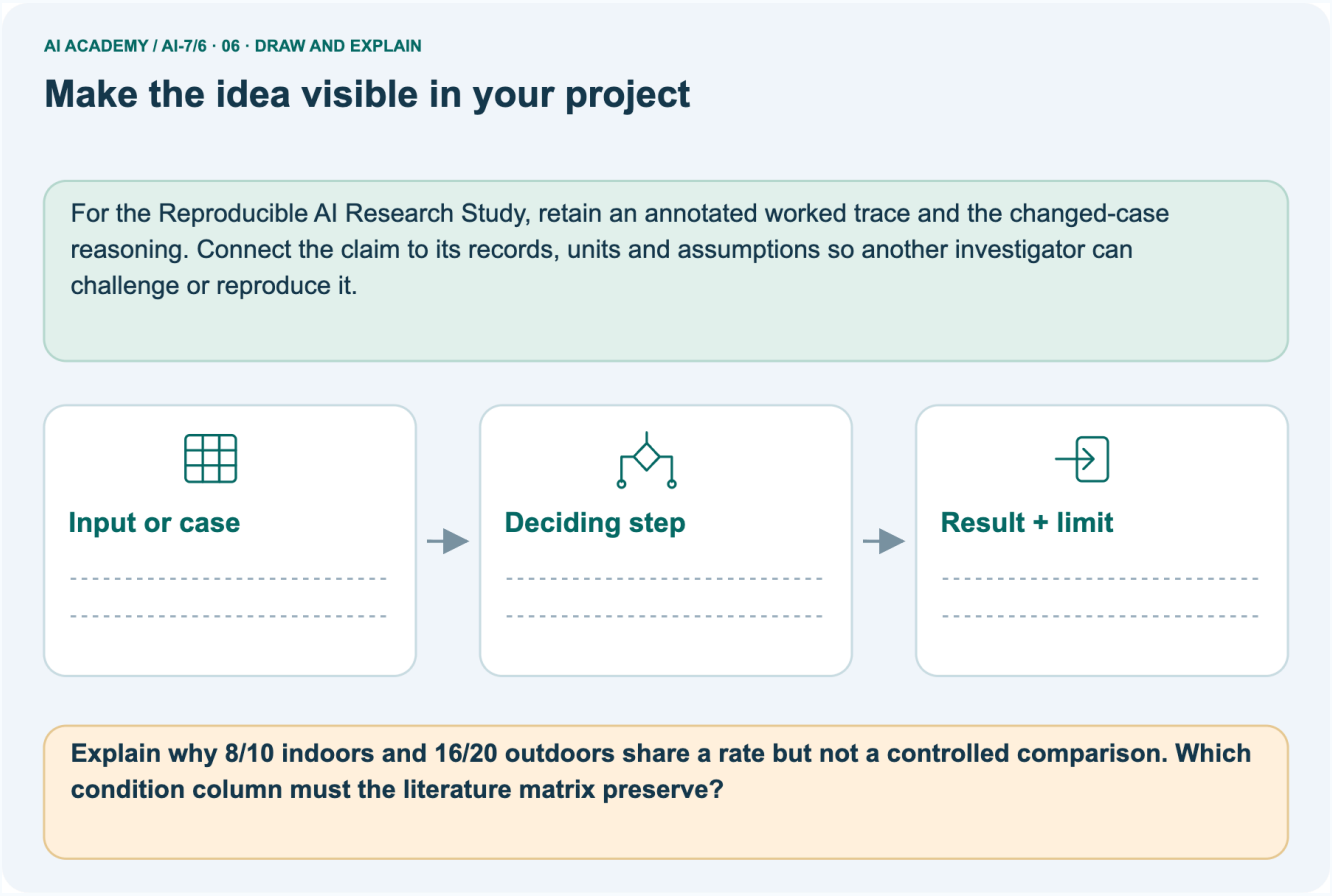
Follow the labelled picture with the matching explanation. Open the vector diagram

A **literature matrix** helps compare **evidence** without flattening differences. Equal percentages do not mean equivalent tasks, samples or methods.

Limit: These fictional studies illustrate comparison practice; no real literature finding is asserted.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the **literature matrix** preserve?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. A literature comparison matrix: the case

02. Follow the deciding steps

03. Same rate, different study conditions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.